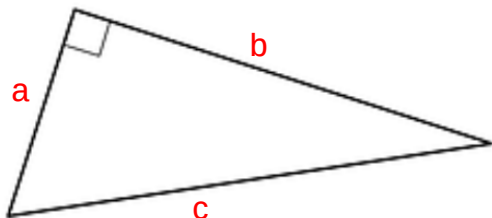


Grade 8
Unit 5
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

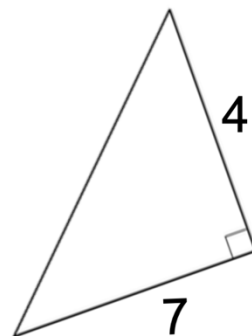
1. For the right triangle below, label the legs a and b , and the hypotenuse c .



Use the right triangle with side length 4 and 7 for questions 2–5.

2. Which side length is unknown?

☐ Leg
☒ Hypotenuse



3. In the equation for the Pythagorean Theorem, $a^2 + b^2 = c^2$, substitute the given values for

☒ a and b
☐ a and c
☐ b and c

4. Substitute the values in the equation and algebraically solve for the missing variable.

$$a^2 + b^2 = c^2$$

$$4^2 + 7^2 = c^2$$

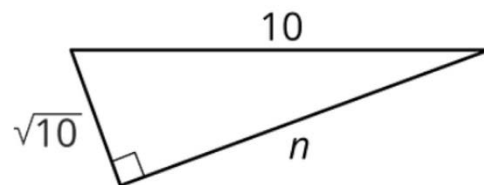
$$16 + 49 = c^2$$

$$65 = c^2 \qquad c = \sqrt{65}$$

5. Identify the two solutions.

Exact Solution	Approximate Solution
$c = \sqrt{65}$	$c \approx \underline{8.062}$

Use the right triangle with side length 10 and $\sqrt{10}$ for questions 6–10.



6. Levi wrote the equation $10^2 + b^2 = \sqrt{10}^2$ to find the missing side length, but made a mistake. What mistake did he make?

c , or the hypotenuse, measures 10, and the leg, or a , measures $\sqrt{10}$.

7. Which side length is unknown?

- ☒ Leg
☐ Hypotenuse

8. State the Pythagorean Theorem and substitute the given values for

$$a^2 + b^2 = c^2$$

$$(\sqrt{10})^2 + b^2 = 10^2$$

- ☐ a and b
☒ a and c
☐ b and c

could also use b and c depending on how you label the triangle.

9. Solve the equation for the missing variable.

$$(\sqrt{10})^2 + b^2 = 10^2$$

$$10 + b^2 = 100$$

$$b^2 = 90$$

$$b = \sqrt{90}$$

10. Identify the two solutions.

Exact Solution	Approximate Solution
$b = \sqrt{90}$	$b \approx \underline{9.487}$